

SQL Quick Reference: Commands & Administration

A comprehensive reference guide for Basic Data Manipulation (DML/DQL), Extra Optimization Tools, and Database Management/Metadata Commands.

1. Metadata & Management Commands (Show/Inspect Options)

These utility commands are crucial for inspecting your system parameters, viewing database environments, examining schemas, and identifying configuration rules.

Command & Description	Example Syntax
SHOW DATABASES Lists all available databases in the server instance.	<code>SHOW DATABASES;</code>
USE Switches the context to a specific database environment.	<code>USE company_db;</code>
SHOW TABLES Displays all existing tables in the currently active database.	<code>SHOW TABLES;</code>
DESCRIBE / DESC Inspects column attributes, data types, and primary key rules of a table.	<code>DESCRIBE employees;</code>
SHOW COLUMNS An alternative to describe, extracting specific structure layouts.	<code>SHOW COLUMNS FROM orders;</code>
SHOW VARIABLES Inspects internal engine settings, timeout thresholds, and global configurations.	<code>SHOW VARIABLES LIKE '%timeout%';</code>
SHOW PROCESSLIST Lists active queries running right now to identify locked or slow connections.	<code>SHOW PROCESSLIST;</code>

2. Basic Commands (Core Essentials)

Standard syntax structures for retrieving, storing, modifying, and filtering rows.

Command	Example Syntax
SELECT / FROM Queries database fields.	<code>SELECT first_name, salary FROM employees;</code>
WHERE Applies conditions to filter results.	<code>SELECT * FROM employees WHERE dept = 'IT';</code>
INSERT INTO Appends new records.	<code>INSERT INTO employees (id, name) VALUES (1, 'Sam');</code>
UPDATE Modifies text/values on a filtered subset.	<code>UPDATE employees SET status = 'Active' WHERE id = 1;</code>
DELETE Erases individual rows safely.	<code>DELETE FROM employees WHERE status = 'Terminated';</code>

3. Extra & Analytical Commands

Performance assessment, safe query bundling, structure adjustments, and formatting tools.

Command	Example Syntax
ORDER BY Arranges results sequentially (ASC/DESC).	<code>SELECT * FROM items ORDER BY price DESC;</code>
GROUP BY / HAVING Calculates aggregations and filters them.	<code>SELECT role, AVG(pay) FROM staff GROUP BY role HAVING AVG(pay) > 40000;</code>
ALTER TABLE Adds or alters database columns dynamically.	<code>ALTER TABLE clients ADD phone_number VARCHAR(15);</code>
EXPLAIN Visualizes database query execution plans.	<code>EXPLAIN SELECT * FROM orders WHERE user_id = 99;</code>
COMMIT / ROLLBACK Manages atomic financial transaction safety states.	<code>BEGIN; UPDATE ledger SET cash = 50; COMMIT;</code>

Generated automatically as a structural reference sheet. Keep syntax variance for alternate engines (e.g., PostgreSQL vs MySQL) in mind.